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COMPLETE
SPECIFICATION
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**ONLINE SERVICE BOOKING PLATFORM WITH PAYMENT
INTEGRATION AND AI DRIVEN PROVIDER MATCHING**

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The following specification particularly describes the invention and the way it is to be performed.

AN AI-POWERED CONTEXT-AWARE OUTFIT RECOMMENDATION SYSTEM USING STYLE, SEASON, AND OCCASION INPUTS

FIELD OF INVENTION:

This invention relates to Artificial Intelligence and Machine Learning (AI/ML) Engineering.

This invention pertains to context-aware recommendation systems and human-centered computing applications in the domain of fashion technology. The proposed system focuses on the design and implementation of an AI-powered outfit recommendation system that generates personalized clothing recommendations based on contextual inputs such as style preferences, season, weather conditions, occasion, and user mood.

BACKGROUND OF INVENTION:

The fashion industry has undergone significant digital transformation with the rapid growth of e-commerce platforms and online retail services. Consumers today are exposed to an extensive variety of clothing and accessories, making outfit selection increasingly complex and time-consuming. Fashion is not only a functional necessity but also a medium of personal expression, identity, and social communication. As a result, users demand intelligent systems that can assist them in selecting outfits that align with their personal style, current trends, and situational needs.

Traditional fashion recommendation systems primarily rely on historical user behavior, purchase patterns, or content-based similarity using product images. Many existing systems employ machine learning and deep learning techniques such as Convolutional Neural Networks (CNNs), K-Nearest Neighbors (KNN), and pre-trained architectures like ResNet to extract visual features and recommend visually similar clothing items. While these approaches have demonstrated effectiveness in image-based similarity matching and product classification, they often lack awareness of real-world contextual factors such as season, weather conditions, occasion, or user mood. Consequently, recommendations generated by such systems may be visually relevant but contextually inappropriate.

Recent research highlights the importance of incorporating contextual intelligence into recommendation systems to improve relevance and user satisfaction. Factors such as time, location, climate, and social context play a crucial role in clothing decisions, yet they are frequently ignored or underutilized in existing fashion recommender models. Furthermore, many systems depend heavily on cloud-based architectures, which can introduce latency, privacy concerns, and limited personalization in low-connectivity environments.

Advancements in artificial intelligence, particularly the emergence of large language models (LLMs), have enabled more natural and human-like interaction between users and intelligent systems. However, the integration of locally deployed LLMs with fashion recommendation platforms remains largely unexplored. Combining contextual reasoning, natural language understanding, and visual recommendation capabilities presents an opportunity to move beyond static product similarity toward more personalized and adaptive outfit suggestions.

Therefore, there exists a clear need for an AI-powered, context-aware fashion recommendation system that integrates machine learning, contextual inputs, and natural language intelligence to generate personalized outfit recommendations. The present invention addresses these limitations by proposing a system that considers style preferences, season, weather, occasion, and user mood while delivering visually rich and interactive outfit suggestions through a user-friendly web-based interface.

OBJECTIVE OF INVENTION:

The objective of the present invention is to develop an AI-powered contextual outfit recommendation system that generates personalized clothing suggestions based on user preferences, season, weather, and occasion. The system aims to improve recommendation accuracy using machine learning techniques while providing a user-friendly and visually appealing interface that simplifies outfit selection and enhances the overall user experience.

SUMMARY OF INVENTION:

The present invention discloses an AI-based contextual outfit recommendation system designed to assist users in selecting suitable clothing combinations based on multiple contextual and personal factors. The system analyzes user inputs such as style preferences, season, weather conditions, and occasion to generate personalized outfit recommendations. Machine learning algorithms are employed to identify patterns in user preferences and contextual data, thereby improving the relevance and accuracy of the suggested outfits.

The system provides a web-based, user-friendly interface that displays visually appealing outfit inspirations retrieved from online image sources. Users can easily view, regenerate, and save recommended outfits for future reference. The modular and scalable architecture of the invention allows for easy expansion with additional contextual parameters or recommendation techniques, making it adaptable to diverse user needs.

DETAIL DESCRIPTION:

The system relates to an AI-powered fashion recommendation system designed to provide personalized outfit suggestions based on multiple contextual inputs, including style preferences, season, weather conditions, and user-specific occasions. The system leverages machine learning algorithms, particularly Convolutional Neural Networks (CNN), ResNet50, and K-Nearest Neighbor (KNN), to process images of clothing items and generate recommendations that align with user preferences and contextual parameters.

The system comprises three primary modules: Input and Context Processing, Training and Feature Extraction, and Recommendation Generation.

1. Input and Context Processing

The system accepts user inputs, including preferred style (e.g., casual, formal, sporty), season (e.g., summer, winter), occasion (e.g., party, office, casual outing), and optionally weather data (e.g., temperature, precipitation). In addition, the system allows uploading or selecting a reference outfit image for visual similarity matching.

Images are pre-processed by normalizing pixel values and resizing them to a uniform resolution (e.g., 128×128 pixels). Random image transformations, such as horizontal flips and minor rotations, are applied to enhance the generalization capability of the model. Orientation metadata is also used to correctly align the images for consistent feature extraction.

2. Training and Feature Extraction

The system uses a labeled fashion image dataset, containing images annotated with attributes such as color, pattern, article type, and contextual metadata. The dataset is divided into training and testing subsets, typically with an 80%-20% split.

The CNN model, based on ResNet50 architecture, extracts high-dimensional feature vectors from each image. The ResNet50 model consists of 50 layers, including convolutional layers, batch normalization layers, activation functions (ReLU), pooling layers, and residual connections. Residual connections mitigate the vanishing gradient problem, enabling deeper network training.

The feature extraction process encodes key visual aspects such as color, texture, pattern, and garment shape. These feature vectors are stored in a searchable database, linked to their corresponding contextual attributes.

3. Recommendation Generation

For a given user query, the system retrieves relevant images and computes similarity scores between the query and database items using Euclidean distance in the feature space. The K-Nearest Neighbor (KNN) algorithm identifies the top-k items with the closest feature vectors.

Additionally, the system filters recommendations based on contextual inputs:

Season: Outfits are filtered to match seasonal appropriateness (e.g., coats for winter, dresses for summer).

Occasion: Items are selected according to user-specified occasion tags (e.g., formal shirts for office, party dresses for events).

Style Preferences: Recommendations prioritize items matching the preferred style profile of the user.

The recommended outfits are then displayed in a mobile-friendly, visually appealing interface, which supports carousel views, theme toggling, and bookmarking of favorite outfits. Users can regenerate recommendations dynamically based on updated inputs

WE CLAIM

1. An AI-powered context-aware outfit recommendation system, comprising:
 - a user interface configured to receive user inputs including style preference, season, occasion, and optional weather condition;
 - an image processing module adapted to preprocess fashion images;
 - a feature extraction module based on a deep learning architecture selected from Convolutional Neural Networks or ResNet models;
 - a recommendation engine employing a similarity-based algorithm selected from K-Nearest Neighbors or equivalent models; and
 - a presentation module configured to display personalized outfit recommendations, wherein the system generates contextually appropriate outfit recommendations by jointly analyzing visual features and user-provided contextual inputs.
2. The system as claimed in claim 1, wherein the contextual inputs include season-based and weather-based constraints to ensure climate-appropriate outfit recommendations.
3. The system as claimed in claim 1, wherein the recommendation engine filters clothing items based on occasion-specific metadata including formal, casual, party, or professional usage.
4. The system as claimed in claim 1, wherein the feature extraction module utilizes a pre-trained ResNet50 architecture to encode visual attributes including color, texture, pattern, and garment shape.
5. The system as claimed in claim 1, wherein similarity between clothing items is computed using Euclidean distance or cosine similarity within a high-dimensional feature space.
6. The system as claimed in claim 1, wherein the system supports interactive features including dynamic regeneration, outfit preview, bookmarking, and storage of user preferences.

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